

SPECTRAL GRAPH THEORY AND THE MATRICES $J_n(\bar{j})$, $H_n(\bar{h})$: REMARKS ON THE DIGRAPH VIEWPOINT

These remarks concern the two matrices defined after Lemma 2.1 of *Oscillating spectra associated to certain sequences*, regarded as weighted adjacency matrices of directed graphs. The short answer is yes: spectral graph theory says quite a bit, and the directed-graph viewpoint fits these matrices unusually well. The remarks are organized from “structural” to “potentially useful for the μ_n project.” References to proofs of every result invoked are given inline, and Section 7 is an annotated guide to the bibliography intended for a reader with little background in spectral graph theory.

1. THE DIGRAPH OF J_n HAS A VERY RIGID CYCLE STRUCTURE

Read $J_n(\bar{j})$ as the weighted adjacency matrix of a digraph on vertices $1, \dots, n$, with the convention that the entry A_{uv} is the weight of the edge $u \rightarrow v$. (For the general dictionary between square matrices and weighted digraphs, see Brualdi and Cvetković [4], the gentlest introduction we know, or Chapter 6 of Horn and Johnson [8].) The superdiagonal gives a single “ascending” path $i \rightarrow i+1$ with weight $-i$, and the lower-triangular Toeplitz part gives “descending” edges $i \rightarrow k$ ($k \leq i$) with weight j_{i-k+1} , including loops of weight j_1 . Because the *only* way to ascend is one step at a time, every directed cycle is supported on an interval $[k, m]$: it climbs $k \rightarrow k+1 \rightarrow \dots \rightarrow m$ and returns via the single edge $m \rightarrow k$. Its weight is

$$(1) \quad w([k, m]) = j_{m-k+1} (-1)^{m-k} \frac{(m-1)!}{(k-1)!}.$$

2. THE COEFFICIENT THEOREM FOR DIGRAPHS REPROVES THEOREM 2.3 COMBINATORIALLY

The *coefficient theorem* for weighted digraphs — in the graph-theory literature it is usually attributed to Sachs and appears, stated and proved for digraphs, in Chapter 1 of Cvetković, Doob, and Sachs [5] — says that the coefficient of x^{n-r} in $\chi(J_n)$ is

$$\sum_L (-1)^{c(L)} w(L),$$

the sum running over collections L of vertex-disjoint cycles covering r vertices, where $c(L)$ is the number of cycles in L and $w(L)$ is the product of the edge weights occurring in L . (The theorem is an unpacking of the Leibniz expansion of $\det(xI - A)$ via the cycle decomposition of permutations. Self-contained proofs accessible without graph-theoretic background may be

found in Brualdi–Cvetković [4], where determinants and characteristic polynomials are developed from the start in the language of weighted digraphs, and in Stanley [11, §4.7], as the first step of the transfer-matrix method.)

By the previous section, the relevant collections L are exactly sets of pairwise disjoint intervals in $\{1, \dots, n\}$ — that is, the coefficients are signed sums over partial composition-like structures, which is morally why partition and composition identities (MacDonald, Vein–Dale) appear at all in this setting.

This was verified numerically (see the script in Section 8): for a random integer sequence $\bar{j}$ with $j_1 = 1$ and $n = 7$, the cycle-cover sums match both `numpy`’s characteristic polynomial and the formula of Theorem 2.3,

$$\binom{n}{r} r! h_r (-1)^r,$$

exactly.

A corollary worth noting: for digraphs the spectrum depends *only* on the multiset of cycle products. One direction is immediate from the coefficient theorem, since the coefficients of χ are assembled from cycle products alone. The deeper converse-type statement — two matrices with the same digraph are diagonally similar if and only if all corresponding cycle products agree — is a theorem of Saunders and Schneider [10]; diagonal similarity is the matrix form of a “gauge transformation” that reweights edges while preserving cycle products. Hence the interval weights (1) are the complete intrinsic data behind the spectra under study. Any answer to the question “what is it about $\chi(J_n)$ that induces the observed oscillations?” must be expressible in terms of them.

3. STRONG CONNECTIVITY AND ITS STANDARD DIVIDENDS

The digraph is strongly connected (one can ascend stepwise and descend in one hop), so J_n is irreducible: the equivalence “irreducible $\Leftrightarrow$ strongly connected digraph” is proved in [8, Ch. 6] and in [1, Ch. 2]. Moreover, since the superdiagonal is nowhere zero, J_n is an *unreduced* Hessenberg matrix, hence nonderogatory — every eigenvalue has geometric multiplicity 1, and $\chi(J_n)$ coincides with the minimal polynomial. (Proof in one line: deleting the first column and last row of $J_n - \lambda I$ leaves an upper-triangular matrix with diagonal $-1, -2, \dots, -(n-1)$, which is nonsingular, so $\text{rank}(J_n - \lambda I) \geq n-1$ for every λ . The standard treatment of unreduced Hessenberg matrices is Golub and Van Loan [7, §7.4].)

Irreducibility upgrades the Geršgorin disc theorem [6] via Taussky’s theorem: an eigenvalue lying on the boundary of the union of the discs must lie on the boundary of *every* disc. Taussky’s original article is [12]; modern proofs of both the disc theorem and Taussky’s refinement appear in [8, Ch. 6] and, with much more leisurely exposition, in Chapter 1 of Varga’s book [13], which is devoted entirely to this circle of ideas and is written to be readable without prior background. More interestingly, Brualdi’s cycle-based

refinement of Geršgorin is spectral graph theory tailor-made for the present situation: it localizes eigenvalues using products of deleted row sums taken *around directed cycles* of the digraph, and here the cycles are exactly the intervals described above. Brualdi's theorem, with proof, is in his original paper [3]; a textbook proof, together with comparisons against the classical discs and Brauer's ovals of Cassini, is in Chapter 2 of [13].

4. THE GERŠGORIN PICTURE EXPLAINS, GEOMETRICALLY, WHAT DEFORMATION DOES

Every diagonal entry of J_n equals j_1 , so all Geršgorin discs share one center. Deformation by c replaces every loop weight by c (and shifts the descending-edge weights), so it translates the common disc center to c while reorganizing the radii. In graph language, deformation is a uniform reweighting of the loops — the spectrum is pinned to circulate around c . This gives at least a heuristic for why the deformed spectra look more regular: the eigenvalues live in an annulus-like region about a fixed center, and μ_n measures the distance from 0 to a cloud orbiting c .

5. A CONCRETE LEVER FOR BOUNDING μ_n AWAY FROM ZERO

Since the determinant is the product of the eigenvalues (with multiplicity),

$$\prod_i |\lambda_i| = |\det J_n| = n! |h_n|,$$

any upper bound ρ on the spectral radius coming from the graph (Geršgorin row sums in terms of the j 's — see [8, Ch. 6] or [13, Ch. 1] — or Brualdi's cycle bounds [3]) yields

$$\mu_n \geq \frac{n! |h_n|}{\rho^{n-1}}.$$

This is exactly the shape of statement the project wants — it reduces “the spectrum is zero-free” to controlling the top of the spectrum graph-theoretically, plus knowing $h_n \neq 0$. For the Lehmer application the logic unfortunately runs the wrong way ($h_n \neq 0$ is the desired conclusion, not a hypothesis). Still, for the deformed matrices, where only zero-freeness is needed, it is potentially usable.

6. CAVEATS, AND THE MATRIX H_n

One important caveat: the classical heavy machinery of spectral graph theory assumes nonnegative or symmetric matrices. The Perron–Frobenius theorem (proved in [8, Ch. 8], in [1, Ch. 2], and — in what is perhaps the most readable account — in Chapter 8 of Meyer [9]) requires nonnegative entries; eigenvalue interlacing and its many graph-theoretic consequences (see [8, §4.3] for the matrix statement, or Brouwer and Haemers [2] for the spectral-graph-theory applications) require symmetry. The weights here are signed and the matrices are far from normal, so the spectra are genuinely

complex point sets and those tools do not apply. What survives is exactly the digraph toolkit above: the coefficient theorem, irreducibility, and cycle-based inclusion regions.

Everything said for J_n transfers to $H_n(\vec{h})$ with the obvious changes: the superdiagonal weights are all $+1$, and the first-column edges $i \rightarrow 1$ carry weight $i h_i$. Consequently the cycles of the H_n digraph are the loops, the intervals $[k, m]$ with $k \geq 2$, and the intervals $[1, m]$, the last carrying the extra factor m .

7. GUIDE TO THE REFERENCES

For a reader new to spectral graph theory we suggest the following order. References below are given at the chapter or section level rather than by theorem number, since numbering differs between editions.

Start here. Brualdi–Cvetković [4] develops elementary matrix theory (determinants, characteristic polynomials, Cayley–Hamilton) entirely through weighted digraphs and cycle covers, assuming only linear algebra; it contains everything needed for Sections 1–2 of these remarks, with proofs. Stanley [11, §4.7] gives a compact alternative proof of the cycle-cover expansion as part of the transfer-matrix method.

The eigenvalue-inclusion world. Varga [13] is a short, self-contained book devoted to Geršgorin-type theorems: Chapter 1 proves the disc theorem and Taussky’s boundary refinement, and Chapter 2 proves Brualdi’s cycle-based inclusion sets used above. The original sources are Geršgorin [6], Taussky [12], and Brualdi [3]; Brualdi’s paper is also the best place to see how the digraph cycle structure enters the proofs.

Standard references. Horn–Johnson [8] (already cited in the main paper for Proposition 2.4 there) contains proofs of every matrix-theoretic fact we used: Geršgorin, Taussky, and irreducibility in Chapter 6, interlacing in §4.3, Perron–Frobenius in Chapter 8. Golub–Van Loan [7, §7.4] covers unreduced Hessenberg matrices. Berman–Plemmons [1] and Meyer [9] are alternatives for Perron–Frobenius and irreducibility, Meyer being the gentlest.

Spectral graph theory proper. Cvetković–Doob–Sachs [5] is the classical monograph; its Chapter 1 contains the coefficient theorem for digraphs with proof. Brouwer–Haemers [2] (freely available from the authors’ web pages) is a modern, readable account of the symmetric/undirected theory — useful background, with the caveat that most of it does not apply to the signed, nonsymmetric matrices studied here. Saunders–Schneider [10] proves the equivalence between diagonal similarity and equality of cycle products.

8. THE PYTHON VERIFICATION SCRIPT

The following script (i) builds $J_n(\vec{j})$ for a random integer sequence with $j_1 = 1$; (ii) computes its characteristic polynomial with `numpy`; (iii) recomputes the coefficients from the digraph coefficient theorem, enumerating all sets of pairwise disjoint intervals and using the cycle weights (1); and (iv)

compares both against the formula of Theorem 2.3, with h recovered from j via equation (D1). All three computations agree.

```
import numpy as np
import random
from math import comb, factorial

random.seed(1)
n = 7
# random integer j-sequence with j_1 = 1 (as in Lemma 2.2)
j = [None, 1] + [random.randint(-4, 4) for _ in range(2, n + 1)] # j[1..n]

# Build J_n(jbar): lower part Toeplitz j_{i-k+1}, superdiagonal -i
J = np.zeros((n, n))
for i in range(1, n + 1):
    for k in range(1, n + 1):
        if k <= i:
            J[i - 1, k - 1] = j[i - k + 1]
        elif k == i + 1:
            J[i - 1, k - 1] = -i

# Characteristic polynomial via numpy:
# coefficients of x^n + a_1 x^{n-1} + ...
cp_numeric = np.poly(J)

# --- digraph coefficient theorem ---
# Cycles of the digraph = intervals [k,m]:
# weight = j_{m-k+1} * (-1)^{(m-k)} * (m-1)!/(k-1)!
def cycle_weight(k, m):
    return j[m - k + 1] * ((-1) ** (m - k)) * factorial(m - 1) // factorial(k - 1)

# Coefficient of x^{n-r}: sum over sets of disjoint intervals
# covering r vertices, with sign (-1)^{#cycles} (det(xI - A) convention)
intervals = [(k, m) for k in range(1, n + 1) for m in range(k, n + 1)]

def disjoint_sets(ivs):
    """Enumerate all sets of pairwise disjoint intervals.
    """
    out = [[]]
    def rec(start, chosen):
        for idx in range(start, len(ivs)):
            iv = ivs[idx]
```

```

        if all(iv[0] > c[1] or iv[1] < c[0] for c in
            chosen):
            chosen.append(iv)
            out.append(list(chosen))
            rec(idx + 1, chosen)
            chosen.pop()
    rec(0, [])
    return out

coeffs = [0] * (n + 1) # coeffs[r] multiplies x^{n-r}
coeffs[0] = 1
for S in disjoint_sets(intervals):
    if not S:
        continue
    r = sum(m - k + 1 for k, m in S)
    w = 1
    for k, m in S:
        w *= cycle_weight(k, m)
    coeffs[r] += ((-1) ** len(S)) * w

print("numpy_char_poly_coeffs: ", np.round(cp_numeric,
    6))
print("digraph_cycle-cover_coeffs:", coeffs)
print("match:", np.allclose(cp_numeric, coeffs))

# --- confirm these equal Theorem 2.3's binom(n,r) r! h_r
#      (-1)^r,
#      with h recovered from j via equation (D1) ---
h = [1]
for m in range(1, n + 1):
    s = sum(j[r] * h[m - r] for r in range(1, m + 1))
    h.append(s / m)
thm = [comb(n, r) * factorial(r) * h[r] * (-1) ** r for r
    in range(n + 1)]
print("Theorem_2.3_coeffs: ", thm)

```

Output for the seeded run ($n = 7$):

```

numpy char poly coeffs:      [ 1. -7. -21. 385. -805.
    -609. 6433. -8041.]
digraph cycle-cover coeffs: [1, -7, -21, 385, -805, -609,
    6433, -8041]
match: True
Theorem 2.3 coeffs:         [1, -7.0, -21.0, 385.0,
    -805.0, -609.0,
                                6433.0, -8041.0]

```

REFERENCES

- [1] A. Berman and R. J. Plemmons, *Nonnegative Matrices in the Mathematical Sciences*, Classics in Applied Mathematics 9, SIAM, Philadelphia, 1994. [Chapter 2 proves the Perron–Frobenius theorem and the equivalence of irreducibility with strong connectivity of the digraph.]
- [2] A. E. Brouwer and W. H. Haemers, *Spectra of Graphs*, Universitext, Springer, New York, 2012. [A readable modern account of spectral graph theory for symmetric (undirected) adjacency matrices, including Perron–Frobenius and eigenvalue interlacing with proofs; freely available from the authors’ web pages.]
- [3] R. A. Brualdi, Matrices, eigenvalues, and directed graphs, *Linear and Multilinear Algebra* **11** (1982), 143–165. [Original proof of the cycle-based refinement of the Geršgorin disc theorem used in these remarks; the digraph structure of the matrix enters the proof directly.]
- [4] R. A. Brualdi and D. Cvetković, *A Combinatorial Approach to Matrix Theory and Its Applications*, Discrete Mathematics and Its Applications, CRC Press, Boca Raton, 2009. [Develops determinants, characteristic polynomials, and the Cayley–Hamilton theorem from scratch in the language of weighted digraphs and cycle covers. The most accessible source for the coefficient theorem; assumes only elementary linear algebra.]
- [5] D. Cvetković, M. Doob, and H. Sachs, *Spectra of Graphs: Theory and Applications*, 3rd ed., Johann Ambrosius Barth, Heidelberg, 1995. [The classical monograph on graph spectra. Chapter 1 states and proves the coefficient theorem for the characteristic polynomial of a weighted digraph, the form used in these remarks.]
- [6] S. Geršgorin, Über die Abgrenzung der Eigenwerte einer Matrix, *Izv. Akad. Nauk SSSR, Ser. Mat.* **7** (1931), 749–754. [The original disc theorem; cited for history — modern proofs are in [8] and [13].]
- [7] G. H. Golub and C. F. Van Loan, *Matrix Computations*, 4th ed., Johns Hopkins University Press, Baltimore, 2013. [Section 7.4 treats unreduced Hessenberg matrices, including the rank argument showing every eigenvalue has geometric multiplicity one.]
- [8] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, Cambridge, 2013. [The standard reference. Chapter 6 proves the Geršgorin disc theorem, Taussky’s boundary refinement, and the equivalence of irreducibility with strong connectivity; §4.3 proves Cauchy interlacing for Hermitian matrices; Chapter 8 proves Perron–Frobenius. This is the same book cited as Horn and Johnson in the main paper.]
- [9] C. D. Meyer, *Matrix Analysis and Applied Linear Algebra*, SIAM, Philadelphia, 2000. [Chapter 8 contains what is perhaps the most readable proof of the Perron–Frobenius theorem, together with the digraph characterization of irreducibility.]
- [10] B. D. Saunders and H. Schneider, Flows on graphs applied to diagonal similarity and diagonal equivalence for matrices, *Discrete Math.* **24** (1978), 205–220. [Proves that two matrices supported on the same digraph are diagonally similar if and only if corresponding cycle products are equal — the precise sense in which the cycle weights (1) are the complete intrinsic spectral data.]
- [11] R. P. Stanley, *Enumerative Combinatorics, Volume 1*, 2nd ed., Cambridge Studies in Advanced Mathematics 49, Cambridge University Press, Cambridge, 2012. [Section 4.7 (the transfer-matrix method) contains a short self-contained proof of the cycle-cover expansion of $\det(I - xA)$, equivalent to the coefficient theorem.]
- [12] O. Taussky, A recurring theorem on determinants, *Amer. Math. Monthly* **56** (1949), 672–676. [Source of the boundary refinement of Geršgorin for irreducible matrices.]
- [13] R. S. Varga, *Geršgorin and His Circles*, Springer Series in Computational Mathematics 36, Springer, Berlin, 2004. [A short book devoted entirely to eigenvalue-inclusion theorems, written to be self-contained. Chapter 1 proves the disc theorem]

and Taussky's theorem; Chapter 2 proves Brualdi's cycle-based inclusion sets. Recommended as the main reference for a reader new to this material.]